Architecture: EHR x wearable fusion digital twin (all data synthetic)

Five layers, read top to bottom and snaking left to right: each box names the file that implements it

1 DATA

run_pipeline.py

orchestrates stages:
synthea, ehr, wearables,
ingest, models, events

config.py

profiles: sample 50 x 7 d,
full 800 x 14 d
ANCHOR 2026-09-30

Synthea

synthetic patients
demographics, dx dates,
lipids, smoking

ehr/build_ehr.py

labs (4 to 8 visits), fasting insulin,
medicines, complications,
5 SNPs and polygenic score

data/<profile>/ehr/*.csv

patients, conditions, labs,
medications, genomics
(genomics labelled SIMULATED)

Synthea Apache 2.0, Java 17 · 5 CSV tables · labs: HbA1c, glucose, insulin, lipids, eGFR, ALT

2 SIMULATION

wearables/derive_params.py

EHR to physiology: Vmx, kp3, k1, m4, kp1,
secretion, renal threshold, drug effects;
calibrated to eAG from HbA1c

wearables/simulate.py

simglucose UVA/Padova ODE + PI beta cell
behaviour: meals, steps, sleep, heart rate,
HRV, SpO2, wrist temperature

wearables/generate.py

per patient run
writes sim_params.csv

raw_wearables, truth

native rates, gaps, dropouts,
under logged meals;
truth = validation only

CGM every 5 min · meal logging p 0.35 to 0.95 · 15 to 120 min dropouts · 1 minute truth kept

3 INGESTION

load_raw_streams

one long format parquet
per patient: metric,
start, end, value, unit

ingestion/align.py: align_to_timeline

nearest reading within 2.5 min
interpolate gaps of 15 min or less
imputed and gap flags; daily metrics

timeseries.parquet

aligned, pre fusion

fuse_with_ehr

labs, conditions, medicines
dated before the day only
test: no future lab leakage

fused_timeline

EHR + wearables
5 min grid

5 minute grid · gaps over 15 min stay missing · hr_missing and glucose_imputed flags · leakage unit test

4 MODELS

features.py and models/ablation.py

102 causal features: glucose dynamics,
insulin and carbs on board, personal history,
physiology, health record; 60 min ablation

models/events.py

LightGBM: P(>180) and P(<70) within 2 h; quantile forecasts +60 and +120 min,
conformal 80% band; thresholds sensitive, balanced, precise;
patient grouped 5 fold CV; per fold boosters kept for SHAP

results/*

events.csv and json, event_risk.parquet,
ablation.csv and json,
event_models/*.txt boosters

patient grouped 5 fold CV · conformal 80% band · paired Wilcoxon ablation · SHAP from booster contributions

5 SERVING

models/events.py: score_stream

rebuild features from history so far
score with held out fold model
no look ahead (tested)

alert rule

risk above threshold now
and 10 minutes earlier
three alert settings

dashboard/app.py (Streamlit)

Triage board ranks the panel; Patient view with SHAP and what if;
Live monitor with alert log; Model results page

tests (13)

leakage, labels,
streaming = batch

Python 3.12 · pytest · Streamlit + Plotly · LightGBM, scikit-learn

Blue = code module Cream dashed = data artifact Grey dashed = orchestration Green arrow = data flow